

THE AI ENGINEER ROADMAP

26 Sections. Every Core Skill. All Free Resources.

Personalized for **Raed** — Final-year B.Tech (AI & Data Science), GATE DA qualified, targeting AI/ML Engineer & Data Scientist roles in the India and Gulf markets.

Source framework: Veeraj Kantilal Gadda's "What you need to do to become a \$300K AI Engineer" (veedaily19.substack.com), built with an AI Engineer at Amazon. Reorganized here into a project-based, mastery-gated study plan — no time-boxing, move on only when you can build the thing, not just explain it.

A note on the "\$300K" framing: that comp band is real but rare, and concentrated in US Big Tech / top AI labs with high experience bars. It's not representative of India or Gulf hiring. The skill map below is genuinely the right one regardless of geography — treat this as an engineering curriculum, not a salary promise. Section 26 ties it back to your actual targets: CAREERMATE and your Pangram-style detector build.

How to Use This Roadmap

This is organized as 26 sequential blocks. Each block lists **what to learn** and the **exact free resources** to learn it from (GitHub repos, official docs, course repos — no paid courses unless you choose to go deeper on your own).

- **Mastery-gated, not time-boxed.** Don't move to the next section on a calendar. Move on when you can rebuild the core idea from scratch without the tutorial open.
- **Project-based.** Every 3–4 sections, stop and build something that uses what you just learned. Suggested checkpoints are marked with a milestone bar.
- **Sections 1–9 are foundation — don't skip them even if some feels basic.** Sections 10–22 are where actual AI engineering differentiation happens. Sections 23–26 are how you convert all of it into interviews and an offer.
- **Anchor project:** CAREERMATE (your agentic recruitment system with an AI interview bot) is a natural home for Sections 10–14 (agents, MCP, orchestration, evaluation, observability) — build those skills directly into it rather than in a disconnected tutorial repo.
- **Second project:** your AI-text-detection build (aimed at the Pangram Labs-style ML engineer profile) is the right home for Sections 4, 7, 13, and 16 — core ML/NLP, embeddings, evaluation, and fine-tuning.

Table of Contents

- | | |
|---------------------------------|--|
| 1. Python Foundation | 2. Git, GitHub and Linux |
| 3. API and Backend Fundamentals | 4. How LLMs Actually Work |
| 5. Prompt Engineering | 6. Structured Outputs and Tool Calling |
| 7. Embeddings and Vector Search | 8–9. RAG: Basic and Advanced |
| 10. AI Agents | 11. Model Context Protocol |
| 12. Orchestration Frameworks | 13. Evaluation |
| 14. Observability | 15. Cost and Latency Optimization |
| 16. Fine-Tuning | 17. Serving and Inference |
| 18. Multimodal | 19. Security and Safety |
| 20. Data Engineering for AI | 21. Cloud and Deployment |
| 22. LLMOps and CI/CD | 23–25. Interview Preparation |
| 26. Portfolio Projects | |

SECTION 1

Python Foundation

Topics to focus on

- Python syntax and best practices
- Functions and modular programming
- Object-Oriented Programming (OOP)
- Type hints and Pydantic
- Async and await
- Iterators and Generators
- Decorators
- Context Managers
- Exception handling and retry logic
- File handling
- Virtual environments
- pip, uv and dependency management
- Logging
- Testing with pytest

Resources

- [Python cheatsheet \(single page reference\)](#)
- [Awesome Python \(curated library index\)](#)
- [Python design patterns in code](#)
- [Algorithms implemented in Python](#)
- [Practice exercises with solutions](#)

Interview prep

- [Tech Interview Handbook](#)
- [NeetCode solutions repository](#)

SECTION 2

Git, GitHub and Linux

Topics to focus on

- Git fundamentals
- Branching strategies
- Merge conflicts
- Pull requests
- GitHub workflows
- GitHub Actions
- Linux file system
- Common terminal commands
- Shell scripting
- SSH

- [Environment variables and secrets](#)

Resources

- [Pro Git \(full free book\)](#)
- [Git Flight Rules — what to do when things go wrong](#)
- [The Art of Command Line](#)
- [Bash guide](#)

SECTION 3

API and Backend Fundamentals

Topics to focus on

- FastAPI
- Request validation with Pydantic
- Authentication and API keys
- Rate limiting
- Background jobs and task queues
- Streaming responses and Server-Sent Events
- WebSockets
- Postgres basics
- Redis and caching
- Docker

Resources

- [FastAPI source and docs](#)
- [FastAPI best practices \(production-focused\)](#)
- [Full stack FastAPI template](#)

Interview prep

- [System Design Primer](#)
- [System Design by Karan Pratap Singh](#)
- [Awesome System Design Resources](#)

■ **MILESTONE PROJECT** — Build a FastAPI backend with an authenticated endpoint, a Postgres model, and a Redis-cached route. This becomes CAREERMATE's API layer.

SECTION 4

How LLMs Actually Work

Topics to focus on

- Tokens and tokenization
- Context windows
- Embeddings
- Transformer intuition: attention, encoder, decoder

- Pretraining vs instruction tuning vs preference tuning
- Temperature, top-p and top-k
- Stop sequences and max tokens
- Logprobs
- Why models hallucinate
- Base models vs chat models vs reasoning models

Resources

- [Hugging Face LLM Course](#)
- [Course source repository](#)
- [Neural Networks: Zero to Hero \(Andrej Karpathy\)](#)
- [Zero to Hero video playlist](#)
- [Zero to Hero code repository](#)
- [Build a Large Language Model From Scratch \(full code\)](#)
- [nanoGPT](#)
- [minGPT \(smaller teaching version\)](#)
- [Awesome LLM \(papers and model index\)](#)
- [Microsoft Generative AI for Beginners \(21 lessons\)](#)

Interview prep

- [Awesome Generative AI Guide](#)
- [LLM Interview Questions](#)
- [Devinterview LLM questions](#)
- [Devinterview NLP questions](#)

This is the theory backbone for your text-detector project — you can't credibly explain why an AI-detection model works without this section solid first.

SECTION 5

Prompt Engineering

Topics to focus on

- System prompts vs user prompts
- Zero-shot and few-shot prompting
- Chain of thought and reasoning prompts
- Role and format constraints
- Delimiters and separating instructions from data
- Prompt templates and variables
- Prompt versioning
- Prompt chaining vs a single call
- Grounding to reduce hallucination
- Prompt caching

Resources

- [Prompt Engineering Guide \(DAIR.AI\)](#)

- [Anthropic prompt engineering docs](#)
- [Anthropic interactive tutorial \(9 chapters\)](#)
- [Claude Cookbooks](#)
- [OpenAI Cookbook](#)
- [Google Gemini Cookbook](#)

SECTION 6

Structured Outputs and Tool Calling

Topics to focus on

- JSON mode and structured output
- Schema definition with Pydantic
- Function calling and tool use
- Writing tool schemas and tool descriptions
- Parallel tool calls
- Handling malformed output
- Validation and repair loops
- Deterministic fallbacks when the model fails

Resources

- [Instructor \(structured output with Pydantic\)](#)
- [Outlines \(constrained generation\)](#)
- [Guidance](#)

■ **MILESTONE PROJECT** — Add a tool-calling interview-bot skill to CAREERMATE: a schema-validated function the model calls to score or log a candidate answer, with a repair loop for malformed JSON.

SECTION 7

Embeddings and Vector Search

Topics to focus on

- What embeddings actually represent
- Embedding models and dimensions
- Cosine similarity and distance metrics
- Approximate nearest neighbor search: HNSW and IVF
- Vector stores: pgvector, Chroma, Qdrant, Pinecone, Weaviate
- Metadata filtering
- Hybrid search with BM25 plus vectors
- Index tuning and the recall vs latency tradeoff

Resources

- [Sentence Transformers](#)
- [FAISS](#)

- [pgvector](#)
- [Qdrant](#)
- [Qdrant examples](#)
- [Chroma](#)
- [Weaviate](#)
- [Weaviate recipes](#)
- [Pinecone examples](#)
- [VectorHub \(vector retrieval learning hub\)](#)

SECTION 8-9

RAG — Basic and Advanced

Topics to focus on

- **Basic:** document ingestion pipelines; parsing PDFs, HTML, tables and images; chunking strategies (fixed, recursive, semantic, structural); chunk size/overlap tradeoffs; metadata design; retrieval and top-k selection; context assembly and token budgeting; citations and source attribution; handling 'answer not in documents'.
- **Advanced:** query rewriting and expansion; multi-query retrieval; reranking with cross encoders; hybrid retrieval and reciprocal rank fusion; parent document / small-to-big retrieval; contextual retrieval; Graph RAG; agentic RAG; incremental reindexing and document sync; diagnosing retrieval vs generation failures.

Resources

- [RAG Techniques by Nir Diamant \(best free collection, notebook per technique\)](#)
- [RAG From Scratch \(LangChain, video + notebooks\)](#)
- [LlamaIndex](#)
- [Haystack](#)
- [RAGFlow \(full open source RAG engine, reference implementation\)](#)
- [LightRAG](#)
- [bRAG LangChain \(beginner to advanced, notebooks\)](#)
- [RAG cookbooks by Athina](#)
- [Pathway LLM App templates](#)

■ **MILESTONE PROJECT** — Build a RAG pipeline over your own KTU course notes or resumes dataset. Then upgrade it with reranking + hybrid search — ship the before/after as a comparison in your portfolio.

SECTION 10

AI Agents

Topics to focus on

- The agent loop: plan, act, observe
- The ReAct pattern
- Tool design and tool descriptions
- Memory: short term, long term, episodic
- State management

- Planning and task decomposition
- Multi-agent patterns: supervisor, handoff, swarm
- Human in the loop and approval steps
- Loop limits, budgets and timeouts
- Failure recovery and retries

Resources

- [GenAI Agents by Nir Diamant](#) (simple to multi-agent)
- [Agents Towards Production](#) (same author, focused on shipping)
- [Hugging Face AI Agents Course](#)
- [Agents Course repository](#)
- [Microsoft AI Agents for Beginners](#)
- [smolagents](#)
- [LangGraph](#)
- [AutoGen](#)
- [CrewAI](#)
- [Awesome AI Agents](#)

This is the core of CAREERMATE's interview bot: the plan/act/observe loop, memory across interview turns, and human-in-the-loop review before a candidate is scored. Build it here, not in a throwaway tutorial repo.

SECTION 11

Model Context Protocol (MCP)

Topics to focus on

- MCP architecture: hosts, clients, servers
- Tools, resources and prompts
- Transports: stdio and HTTP
- Building an MCP server
- Building an MCP client
- Authorization
- Security considerations for MCP servers

Resources

- [Official MCP documentation](#)
- [Specification repository](#)
- [Reference servers](#) (best way to learn is by reading these)
- [Hugging Face MCP Course](#)
- [MCP Course repository](#)
- [Microsoft MCP for Beginners](#)
- [Awesome MCP Servers](#)

SECTION 12

Orchestration Frameworks

Topics to focus on

- [LangChain](#)
- [LangGraph](#)
- [LlamaIndex](#)
- Provider SDKs: OpenAI, Anthropic, Google
- [smolagents](#)
- [CrewAI](#)
- [DSPy](#)

Resources

- [LangChain](#)
- [LangChain Academy](#) (free course material)
- [Awesome LangChain](#)
- [DSPy](#)
- [LiteLLM](#) (one interface across 100+ providers)
- [LLM Engineer Handbook](#) (curated stack overview)
- [LLM Engineering course](#) by Ed Donner (8 weeks of notebooks)
- [AI Engineering Hub](#) (project-based tutorials)
- [Awesome LLM Apps](#) (working example applications)

■ **MILESTONE PROJECT** — Rebuild CAREERMATE's agent orchestration in LangGraph (or your framework of choice) with an explicit MCP-based tool layer for resume parsing / job matching.

SECTION 13

Evaluation

Topics to focus on

- Building golden datasets
- Task-specific metrics
- Retrieval metrics: recall, precision, MRR, NDCG
- Generation metrics: faithfulness, answer relevance
- LLM-as-judge and where it fails
- Human evaluation and annotation
- Regression testing for prompts
- Offline vs online evaluation
- A/B testing and user feedback loops
- Evaluation tools: Ragas, DeepEval, promptfoo, LangSmith, Braintrust

Resources

- [Ragas](#)
- [DeepEval](#)
- [promptfoo](#)
- [OpenAI Evals](#)
- [LM Evaluation Harness](#)

Directly relevant to a Pangram-style detector role: precision/recall, ROC-AUC, and a documented golden-set methodology are exactly what that job posting screens for.

SECTION 14

Observability

Topics to focus on

- Tracing and spans
- Token and cost tracking
- Latency: time to first token and total latency
- Error rates and fallback rates
- Prompt and output logging
- PII redaction in logs
- Drift detection

Resources

- [Langfuse](#)
- [Arize Phoenix](#)
- [OpenLLMetry](#)

■ **MILESTONE PROJECT** — Instrument CAREERMATE with tracing + cost/latency dashboards. “I can show you the trace” is a strong interview line few fresher candidates can say.

SECTION 15

Cost and Latency Optimization

Topics to focus on

- Prompt caching
- Semantic caching
- Model routing and cascading (small model first)
- Batching
- Streaming
- Context compression and summarization
- Output length control
- Quantization
- Choosing the right model for the task
- Modeling cost per request

Resources

- [LiteLLM \(routing and fallbacks\)](#)
- [vLLM](#) — read the docs on continuous batching and paged attention

SECTION 16

Fine-Tuning

Topics to focus on

- When RAG or prompting beats fine-tuning
- Supervised fine-tuning (SFT)
- LoRA and QLoRA
- PEFT
- Dataset curation and formatting
- Preference tuning with DPO
- Distillation
- Evaluating a fine-tuned model against the base model
- Total cost of ownership

Resources

- [Unsloth \(fastest path to a first fine-tune on free Colab\)](#)
- [LLaMA Factory](#)
- [PEFT](#)
- [TRL \(includes SFT and DPO\)](#)
- [Axolotl](#)
- [LLM Datasets \(curation guidance\)](#)
- [LLM Course by Maxime Labonne \(strong on fine-tuning\)](#)
- [LLM Finetuning notebooks](#)

Free Colab + Unsloth is genuinely enough to run a real LoRA fine-tune with zero cost — fits your zero-budget constraint for the detector project.

SECTION 17

Serving and Inference

Topics to focus on

- Open weight models and their licenses
- Ollama for local development
- vLLM and TGI
- GPU basics: VRAM, batching, KV cache
- Quantization formats: GGUF, AWQ, GPTQ
- Throughput vs latency
- Self-hosting vs API economics

Resources

- [vLLM](#)
- [Ollama](#)
- [llama.cpp](#)
- [Text Generation Inference](#)
- [BentoML](#)

- [Ray](#)

SECTION 18

Multimodal

Topics to focus on

- Vision inputs and document understanding
- OCR and layout parsing
- Speech to text
- Text to speech
- Image generation
- Multimodal embeddings
- Multimodal RAG

Resources

- [Whisper](#)
- [Docling](#)
- [MinerU](#) (PDF to structured data)
- [Unstructured](#)
- [Hugging Face Cookbook](#) (multimodal RAG recipes)

SECTION 19

Security and Safety

Topics to focus on

- Prompt injection: direct and indirect
- Jailbreaks
- Sensitive information disclosure
- System prompt leakage
- Insecure output handling
- Excessive agency
- Vector and embedding weaknesses
- Supply chain risk: models, adapters, MCP servers, packages
- Guardrails and input/output filtering
- Red teaming
- PII handling and data residency

Resources

- [OWASP Top 10 for LLM Applications 2026](#)
- [OWASP project repository](#) (historical versions)
- [garak](#) (LLM vulnerability scanner)
- [NeMo Guardrails](#)
- [Guardrails AI](#)
- [LLM Guard](#)

- [Awesome LLM Security](#)

■ **MILESTONE PROJECT** — Run a basic prompt-injection red-team pass against CAREERMATE's interview bot (candidates could try to manipulate their own scoring). Document what you found and how you guarded against it.

SECTION 20

Data Engineering for AI

Topics to focus on

- Document parsing at scale
- OCR pipelines
- Deduplication
- Chunk metadata design
- Incremental sync and change data capture
- Permissions-aware retrieval
- Data quality checks
- Pipeline orchestration: Airflow, Prefect, Dagster

Resources

- [Unstructured](#)
- [Docling](#)
- [Airflow](#)
- [AI Tutor RAG system \(real ingestion + retrieval pipeline, read end to end\)](#)

SECTION 21

Cloud and Deployment

Topics to focus on

- Docker
- Kubernetes basics
- Serverless functions
- AWS Bedrock
- Azure AI Foundry
- Google Vertex AI
- Secrets management
- Environment and config management
- Cost monitoring and alerting

Resources

- [Google Cloud Generative AI samples](#)
- [Amazon Bedrock workshop](#)

For Gulf-market roles specifically, AWS and Azure show up most often in job postings (Bayt, NaukriGulf, GulfTalent) — prioritize Bedrock and Azure AI Foundry over Vertex if you have to pick one first.

SECTION 22

LLMOps and CI/CD

Topics to focus on

- Prompt version control
- Eval gates in CI
- Canary and shadow deployments
- Feature flags for model changes
- Rollback strategies
- Model version pinning
- Incident response for AI failures
- Data and prompt lineage

Resources

- [Awesome LLMops](#)
- [MLflow](#)
- [LLM Engineers Handbook](#) (full book code, end-to-end pipeline)

■ **MILESTONE PROJECT** — Add a CI pipeline to CAREERMATE that runs an eval suite (from Section 13) on every prompt change before it merges. This is the single most “senior-looking” thing a fresher can ship.

SECTION 23–25

Interview Preparation

Topics to focus on

- **AI/ML specific:** LLM and NLP concept questions, RAG and agent system-design questions, eval methodology questions.
- **Coding rounds:** DSA fundamentals, pattern-based problem solving.
- **System design rounds:** scalability, tradeoffs, and AI-system-specific design (e.g. ‘design a RAG system for 10M documents’).

Resources

- [Awesome Generative AI Guide](#) (actively maintained, interview banks)
- [LLM Interview Questions](#)
- [Devinterview LLM questions](#)
- [Devinterview NLP questions](#)
- [MLQuestions](#)
- [Machine Learning Interviews](#) (Alireza Dirafzoon)
- [Machine Learning Interview](#) (Khang Pham)
- [Data Science Interviews](#) (Alexey Grigorev)

- [NeetCode solutions](#)
- [Tech Interview Handbook](#)
- [System Design Primer](#)
- [Awesome System Design Resources](#)
- [System Design by Karan Pratap Singh](#)

SECTION 26

Portfolio Projects

Topics to focus on

- A portfolio for AI engineering roles needs at minimum: one RAG system, one agentic system, and one evaluated/fine-tuned model — each deployed, not just notebooks.
- For you specifically: CAREERMATE covers agents + RAG + orchestration if you build Sections 10–14 into it directly. Your text-detector project covers the evaluated-ML-model piece for Pangram-style roles.
- Ship both with a README that states the problem, the architecture decision you made and why, the eval numbers, and known limitations — that combination is what separates a portfolio piece from a tutorial clone.

Resources

- [Awesome LLM Apps \(dozens of complete working apps\)](#)
- [AI Engineering Hub](#)
- [Agents Towards Production](#)
- [LLM Engineers Handbook](#)